# MODEL FOUR STEP PROCESS:

1. Assignment to pools $(\rho, \mathrm{A_i})$.
2. Selection Scoring $(\mathrm{S_i})$
3. Selection - select top $\mathrm{K}$.
4. Visualize – bin and plot.

## 1. Soft Assignment

$$\pi_{ij} \propto e^{\left(-\rho \cdot \frac{(A_i - T_j)^2}{2\sigma^2}\right)}$$

$\pi_{ij}$ → Soft preference of player $i$ for team $j$ before we normalize to probabilities and fill seats
$T_j$ → Team target PoolQ, \rho ->. Assortativity knob
$\sigma$ → Fixed length-scale in ability units (default~0.65)

## 2. Unified Selection Score Equation

$$S_i = A_i + \lambda(B - D)$$

$S_i$ = selection score
$A_i$ = individual performance metric
$\lambda$ = environmental weight
environmental impact
$L_{net} = B - D \quad (B \approx$ benefits, $D \approx$ drawbacks)

### Congestion Equation v1

$(B - D) = -L_C$ (congestion)

$$S_i = A_i - \lambda\, L_C$$

Knockout (sim): $\lambda = 0 \Rightarrow S_i = A_i$ (congestion out of selection)
$L_C$ (hard) → count with $A_j > \theta$
$\theta = \mathrm{med}(\mathrm{A}_i \mid Y_i = 1)$
$L_C$ (smooth, sim default) → LOO mean of $\sigma\left(\gamma\left(A_j - \theta\right)\right)\ j \neq i$

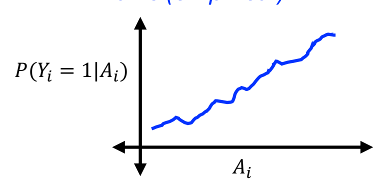
## 3. Selection

The current trivial method is just to choose the top $K$ players by $S_i$
Future methods could introduce stochastic noise $S_i + \delta$

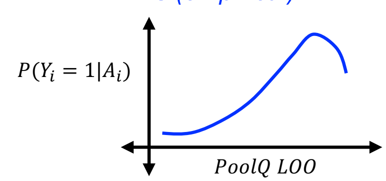
## 4. Visualize

### TALENT vs SELECTION

*Naïve (empirical)*




*HERO (empirical)*




### Empirical (real panel)

Naïve: $P(Y_i = 1|A_i) \rightarrow\ \approx$ monotone     Hero: $P(Y_i = 1|PoolQ\_LOO) \rightarrow$ tail dip